Ifrahuddin Azmi

Embedded and Automation Test Engineer

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Summary

Embedded Software Engineer with 4+ years of experience in developing and **automating firmwaretesting for biosensors**. Specialized in **Python-based automation, BLE/Wi-Fi module testing**, and Linux system administration. Strong problem-solving skills and a collaborative mindset for delivering high- quality embedded solutions.

Technical Skills:

- **Programming**: Python, Shell Scripting, Robot Framework, Selenium, C/C++(Basics)
- **Wireless Protocols:** BLE (Bluetooth Low Energy), Wi-Fi, TCP/IP, LTE, Zigbee
- **Automation Tools:** Robot Framework, TestRail, Jenkins, PyTest
- **Networking Tools:** Wireshark, Scapy, Requests, Socket
- **System Administration:** Linux (Ubuntu, Kali), OpenWrt
- **Version Control & Project Management:** Git, Bitbucket, JIRA
- **DevOps & Systems:** Docker, Kubernetes (basic), GCP, Linux administration (Ubuntu, Kali), cron jobs
- **Web**: Django, FastAPI, Streamlit, Flask
- **Database Management:** NoSQL, MySQL

Education: ,

- **Bachelor of Engineering in Computer Science Engineering** **July 2016 - Jan 2021**
  Visvesvaraya Technological University, Belagavi, India

Professional Experience:

QA Engineer **02/2021 – 03/2022 | BLR**

Transcaal Eng. India Pvt. Ltd.

Diligent Manual Tester with one year of hands-on **experience specializing in Biomedical Devices**, including ECG and Patient monitors, **from March 2021 to March 2022**. Ensured the highest standards of quality assurance in the healthcare technology sector by contributing to rigorous testing and validation processes.

Summary:

Contributed to comprehensive **testing and verification of medical-grade devices**, ensuring compliance with **IEC 62304**
principles for software life cycle processes. Actively participated in d**efect tracking and design- level risk assessment through DFMEA**, helping guarantee the accuracy, reliability, and safety of ECG and Patient monitor functionalities.

Cloud & System Engineering (Part-Time Remote Internship) **09/2023 – 12/2024 | BLR(Remotely)**

Quanterian LLC, Tx – USA

Summary:

Served as a **System Administrator and Cloud Engineer**, specializing in **Google Cloud Platform (GCP).** Key responsibilities included creating and managing instances, deploying applications, and setting up and **monitoring clusters using Kubernetes**. Ensured efficient system performance and scalability while supporting cloud infrastructure remotely from Bangalore (BLR)

Embedded and Automation Test Engineer **03/2022- Present | BLR**
LifeSignals India Pvt. Ltd.

Summary:

- Developed and maintained **automation frameworks for biosensor firmware testing using Python and Robot Framework.**
- Conducted **BLE and Wi-Fi module testing to ensure protocol** compliance and seamless integration.
- Developed **unit test cases and contributed to defect detection and analysis processes**.
- Utilized Wireshark for packet analysis and troubleshooting communication issues.
- Experienced in validating **file systems and protocols like ext4**, NFS, and SMB, focusing on file I/O operations, access control, and storage integration in Linux-based environments.
- Utilized regular expressions **(Regex)** for efficient log parsing, test data validation, and pattern-base
- Built **Python-based scripts for wireless security assessments**, including Deauthentication attacks and fake access points.

Key Achievements:

- Streamlined firmware validation processes by creating reusable automation scripts.
- Enhanced security testing through Python-based Wi-Fi and BLE assessments.

Project 1

Biosensor Firmware Validation

Role Description: In this role, I am responsible **for designing and developing a sophisticated automation system using Robot Framework integrated with Python** to **meticulously test and validate the performance of biosensor firmware.**

Key Responsibilities Include

- Designed automation frameworks using Robot Framework integrated with Python.
- Automated sanity checks and developed Python-based utilities for Git version control.

Project 2

Biosensor Stream Data Security in Python:

Role Description: Proficient Python Developer specializing in biosensor technology, entrusted with the critical responsibility of ensuring data integrity and security

Key Responsibilities Include:

- Implemented **(Regex)** to extract and validate key patterns in biosensor data streams logs
- Developed Python solutions to ensure data validation and integrity for biosensor devices.
- Handled corrupted and null data to ensure uninterrupted functionality and data reliability.
- Simulated real-world scenarios using Scapy for wireless security assessments.
- Utilized Wireshark for in-depth packet analysis, improving network communication robustness.

Project 3

BLE Bleak Automation with Python:

Role Description: Proficient Python Developer specializing in BLE technology, entrusted with the critical responsibility of ensuring data integrity and security.

Key Responsibilities Include:

- Developed Python scripts using the Bleak library to automate BLE communication testing.

- Validated BLE data integrity and implemented solutions for handling corrupted and null data.
- Created Python-based BLE spoofing simulations to assess device security.
- Employed Wireshark for detailed packet analysis of BLE network traffic.
- Collaborated with cross-functional teams to enhance BLE firmware security and reliability.

Project 4

Automated Email Response System:

Description: Developed an automated email response system that integrates with **Office 365** to read, process, and respond to recruiter emails using **Google Generative AI (gemini-1.5-pro-002)**. The solution automates email handling, leveraging **Python** and **AI** for determining when and how to reply to job-related emails, significantly reducing manual effort and ensuring timely responses.

Key Features:

- Created a system to process and respond to recruiter emails using Google Generative AI and Python.
- Automated email retrieval, PDF parsing, and dynamic response generation.

Technical Proficiencies:

- **Python Modules:** bleak, pygatt, socket, subprocess, Scapy, Matplotlib, Pandas
- **Automation Tools:** Robot Framework, Selenium
- **System Tools:** Wireshark, OpenWrt, Linux cron jobs

Wi-Fi Security Skills:

Developing Python Script to Analyze and Enhancing Wi-Fi security:

- **Deauthentication Attacks:**
  - Developing Python scripts for Deauthentication attacks to assess the resilience of Wi-Fi networks against unauthorized disconnection attempts**.**
- **Man-in-the-Middle (MIM) Attacks:**
  - Designing and implementing Python code to simulate Man-in-the-Middle attacks, providing comprehensive testing of network security protocols.
- **Wi-Fi Sniffing and Analysis:**
  - Utilizing Python for Wi-Fi sniffing, capturing and analyzing network traffic to identify vulnerabilities and strengthen overall security.
- **Security Assessments:**
  - Conducting thorough security assessments through code-driven Wi-Fi attacks, ensuring robust protection against potential threats.

Personal Info:

Father's Name	: Mohd Salahuddin
DOB	: 03/02/1999
Nationality	: Indian
Hobbies	: Cricket, Playing PC Game, Badminton, Cooking.
Languages:	: Hindi (Read, write, speak), English (Read, write, speak), Urdu (Read, write, speak), Arabic(read)

Declaration:

I authorize the company or organization to verify the details mentioned in this resume and any other information provided during the recruitment process. I also understand that the data submitted in this resume will be used solely for the purpose of evaluating my suitability for the position I am applying for.

Place: Bengaluru/Karnataka